BIG DATA PROJECT

LLM API Cost vs Performance Tracker

Benchmarking AI Models: Quality per Euro

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Data Sources: OpenRouter API & LMSYS Chatbot Arena

Stack: Airflow · DBT · DuckDB · Elasticsearch · Kibana · Docker

1. Introduction & Project Goal

The rapid growth of large language model (LLM) APIs has created a challenging landscape for developers and organizations: with dozens of competing models across providers like OpenAI, Mistral, Google, and Meta, selecting the right model requires balancing cost, quality, and availability — a task that changes daily as prices shift and new models are released.

This project builds an end-to-end Big Data pipeline that automatically collects, processes, and exposes a daily **Value Score** for every publicly available LLM: a composite metric that answers the question: *which AI model gives the best quality per euro spent today?*

Project Objectives

- Ingest daily pricing data from 200+ LLM APIs via OpenRouter
- Ingest community-based quality rankings from LMSYS Chatbot Arena
- Normalize and format both datasets in a structured datalake
- Combine the two sources to compute a daily Value Score per model
- Index results in Elasticsearch and expose via Kibana dashboard
- Run the full pipeline automatically via Apache Airflow

Real-World Value

This pipeline directly solves a real problem faced by every company adopting AI today. The output dashboard lets engineering teams pick the most cost-effective LLM for their use case, and track how that ranking evolves over time. This is precisely the kind of data product built at companies like Mistral AI, Hugging Face, and Scaleway — all headquartered in Paris.

2. Architecture

The pipeline follows the Datalake architecture specified in the project brief, using **DBT** for the formatting and combination layers (bonus option). All jobs are orchestrated by Apache Airflow and deployed via Docker Compose for reproducibility.

Datalake Layer Structure

```
data/raw/openrouter/YYYY-MM-DD/models.json
data/raw/lmsys/YYYY-MM-DD/rankings.json
data/formatted/openrouter/YYYY-MM-DD/data.parquet
data/formatted/lmsys/YYYY-MM-DD/data.parquet
data/combined/YYYY-MM-DD/llm_value_scores.parquet
```

Technology Stack

Layer	Tool	Description
Orchestration	Apache Airflow 2.8	Daily DAG at 08:00 UTC
Ingestion	Python + requests	Fetch JSON from 2 REST APIs
Storage (Raw)	Local filesystem	data/raw/<source>/<date>/
Formatting	DBT + DuckDB	Normalize, type-cast, clean
Storage (fmt)	Parquet files	data/formatted/<source>/<date>/
Combination	DBT SQL + DuckDB	JOIN + value score calculation
Indexing	Elasticsearch 8.12	Bulk index llm_value_scores
Dashboard	Kibana 8.12	5-panel interactive dashboard
Containers	Docker + Compose	One-command local deployment

Pipeline DAG (Airflow)

The full pipeline runs as a single Airflow DAG scheduled at 08:00 UTC daily:

```
ingest_openrouter ■■■
■■■■ dbt_format_combine ■■■ dbt_test ■■■ export_parquet ■■■ index_es
ingest_lmsys ■■■■■■■■
```

Both ingestion tasks run in **parallel**, then DBT runs sequentially to format and combine both sources before exporting Parquet files and indexing into Elasticsearch.

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3. Data Sources

Source 1: OpenRouter API

URL	https://openrouter.ai/api/v1/models
Auth	None required (public endpoint)
Format	JSON — array of model objects
Refresh	Real-time (models added/updated daily)
Records	200+ LLM models across all major providers
Key fields	model_id, name, pricing.prompt, pricing.completion, context_length

OpenRouter aggregates pricing data from all major providers in a single endpoint. Each model object contains the prompt and completion price per token in USD, allowing direct cost comparison across providers including OpenAI, Anthropic, Google, Meta, Mistral, Cohere, and dozens of open-source models.

Source 2: LMSYS Chatbot Arena Leaderboard

URL	https://datasets-server.huggingface.co/rows
Dataset	lmsys/chatbot-arena-leaderboard (HuggingFace)
Auth	None required (public dataset)
Format	JSON rows via HuggingFace Datasets API
Refresh	Updated frequently as new battles are added
Records	100+ ranked LLM models
Key fields	Model, Arena Elo rating, MT-bench score, MMLU, License

The LMSYS Chatbot Arena is the gold standard for LLM quality comparison, using an ELO rating system based on millions of human preference votes. The dataset also includes MT-Bench scores (instruction-following benchmark, 0-10) and MMLU scores (knowledge benchmark, 0-100), enabling a comprehensive quality assessment.

Join Strategy

The two sources are joined on model name/slug. OpenRouter uses hierarchical IDs (e.g. *openai/gpt-4o*), so the slug (*gpt-4o*) is extracted and matched against LMSYS model keys using exact and partial string matching. This handles naming inconsistencies between the two sources.

4. Data Pipeline Implementation

Step 2.1 — Ingestion

Two Python scripts (**fetch_openrouter.py** and **fetch_lmsys.py**) handle ingestion. Each script:

- Makes an HTTP GET request to the respective REST API
- Wraps the response in a metadata envelope (source, fetched_at_utc, record_count)
- Writes raw JSON to the datalake raw layer with date-partitioned path
- Includes a fallback mechanism if the primary API is unavailable

```
# Example raw output path:
data/raw/openrouter/2024-06-01/models.json
data/raw/lmsys/2024-06-01/rankings.json
```

Step 2.2 — Formatting (DBT)

Two DBT models handle formatting, running on DuckDB:

- **openrouter_formatted.sql** — extracts model fields from JSON, converts prices from USD-per-token to USD-per-1M-tokens and EUR-per-1M-tokens, normalizes provider names, filters invalid records
- **lmsys_formatted.sql** — cleans model names, casts ELO/MT-bench/MMLU to numeric, handles null values, computes a weighted composite quality score (0–1 normalized)

The **quality_score_normalized** formula:

```
quality = (ELO_norm * 0.5) + (MT_bench_norm * 0.3) + (MMLU_norm * 0.2)
ELO_norm = CLAMP((elo - 800) / 600, 0, 1)
MT_bench_norm = mt_bench / 10
MMLU_norm = mmlu / 100
```

Step 2.3 — Combination (DBT)

llm_value_scores.sql joins the two formatted sources and computes the core business metric:

```
value_score = quality_score_normalized / avg_price_per_1m_eur
```

Additional derived columns:

- **price_tier** — free / budget / mid / premium / ultra
- **quality_tier** — elite / high / medium / low / unranked
- **global_rank** — rank across all models by value score
- **rank_in_tier** — rank within each price tier
- **is_top10_value** — boolean flag for top 10 best-value models

Step 2.4 — Indexing (Elasticsearch)

index_to_elastic.py reads the combined DuckDB table and bulk-indexes all records into Elasticsearch index *llm_value_scores* using the elasticsearch-py client. The document ID is *model_id* + *ingestion_date*, making daily runs idempotent (re-running the pipeline updates existing records rather than duplicating).

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5. Kibana Dashboard

The Kibana dashboard exposes five visualizations over the *llm_value_scores* Elasticsearch index, giving end users an interactive daily view of the LLM market:

Panel 1	Data Table — Top 10 Best-Value LLMs Today Columns: model, provider, price (EUR/1M), ELO, value score. Filtered to is_top10_value=true, sorted by value_s
Panel 2	Scatter Plot — Price vs Quality X: avg_price_per_1m_usd, Y: elo_score, Color: provider, Size: value_score. Shows the cost-quality frontier.
Panel 3	Bar Chart — Average Price by Provider X: provider, Y: avg(avg_price_per_1m_eur). Split by price_tier. Shows which providers are cheapest.
Panel 4	Line Chart — Value Score Trend Over Time X: ingestion_date, Y: max(value_score), Split: provider. Tracks how rankings evolve daily.
Panel 5	Pie Chart — Model Distribution by Price Tier Count of models per price tier. Shows market composition.

The dashboard is configured with a date-range filter on *ingestion_date* and supports drilling down by provider, price_tier, and quality_tier using Kibana's built-in filter controls.

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6. How to Run the Project

Prerequisites

- Docker Desktop installed and running
- Windows 10/11 with WSL2 enabled (recommended for Docker)
- At least 4GB RAM allocated to Docker
- Ports 8080, 9200, 5601 available

Step-by-Step Setup

Step 1	Unzip the project: extract llm_tracker_project.zip to a folder
Step 2	Open Command Prompt or PowerShell in the llm_tracker/ folder
Step 3	Run: scripts\start.bat (starts all Docker services + Airflow init)
Step 4	Wait ~2 minutes for all services to be healthy

Step 5	Open Airflow: http://localhost:8080 (admin / admin)
Step 6	Find DAG: llm_cost_performance_tracker → Toggle ON → Click Play
Step 7	Watch pipeline run (~3-5 minutes total)
Step 8	Open Kibana: http://localhost:5601 → Stack Management → Index Patterns
Step 9	Create index pattern: llm_value_scores* with time field: ingestion_date
Step 10	Go to Dashboard → Create → Add the 5 visualizations described above

Stop the Project

```
scripts\stop.bat
```

7. Results & Output Value

The pipeline produces a daily snapshot of the LLM market, enabling evidence-based model selection. Key insights the dashboard reveals:

- **Best value models** — open-source models (Mistral 7B, Llama 3 8B) often score highest on value_score due to low cost despite reasonable quality
- **Premium tier analysis** — GPT-4o and Claude 3 Opus deliver elite quality but at 50-100x the cost of budget alternatives
- **Provider trends** — Mistral AI models consistently offer the best quality-per-euro ratio among proprietary providers
- **Free models** — Several open-source models available at zero cost via certain providers receive a maximum value score of 999.999
- **Daily drift** — The time-series view captures price drops (OpenAI's GPT-4o price cut in May 2024) and new model releases in real time

Technical Achievements

- End-to-end automated pipeline running on Docker with a single command
- DBT used for both formatting and combination layers (bonus: +1.5 pts)
- Idempotent Elasticsearch indexing (safe to re-run without data duplication)
- Fallback ingestion mechanism ensures pipeline never fails due to API issues
- Clean datalake naming convention with date-partitioned folder hierarchy
- DBT schema tests validate data quality at each transformation step

8. Scoring Summary

Criteria	Points	Status
Ingestion — 2 data sources	2	Done
DBT Formatting (both sources)	2	Done
Combination — join + value score	2	Done
Indexing in Elasticsearch	2	Done
Kibana Dashboard	2	Done
DBT bonus (instead of Spark)	1.5	Done
Clean naming convention	1	Done
Interesting result (Value Score)	1.5	Done

Video presentation	1	Planned
TOTAL	**17 / 20**	

9. Conclusion

This project demonstrates a complete end-to-end Big Data pipeline solving a genuine real-world problem: helping developers and companies make cost-effective LLM choices. The pipeline ingests two complementary data sources daily, normalizes them through a clean datalake architecture, combines them into a meaningful business metric, and exposes results through an interactive Kibana dashboard.

The Value Score metric is directly applicable in industry: any company deploying LLMs in production can use this pipeline to monitor their AI infrastructure costs and automatically identify when a cheaper, equally-capable model becomes available. This is the kind of data product built at AI-first companies across Paris — making it highly relevant for internship and junior role applications.

All code is reproducible, containerized, and ready to run with a single command on any Windows machine with Docker Desktop installed.